LUIS BURBANO

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EDUCATION

Ph.D. in Computer Science and Engineering *October 2020 - June 2026*
University of California, Santa Cruz, CA, USA *GPA 3.97/4*

- **Advisor**: Prof. Alvaro Cardenas.
- **Thesis title**: Security of autonomous decision-making agents: From control systems to embodied AI.
- **Committee:** Chair: Prof. Alvaro Cardenas, Members: Prof. Cihang Xie, Prof. Daniel Fremont.

M.Sc. in Computer Science and Engineering *December 2025*
University of California, Santa Cruz, CA, USA *GPA 3.97/4*

- **Advisor**: Prof. Alvaro Cardenas.

M.Sc. in Electronics and Computer Engineering *August 2017 - June 2019*
Universidad de los Andes, Colombia *GPA 4.64/5*

- **Advisor**: Prof. Nicanor Quijano.

B.Sc. in Electronics Engineering *January 2013 - June 2017*
Universidad de los Andes, Colombia *GPA 4.26/5*

RESEARCH EXPERIENCE

Visiting Scholar *August 2026 - May 2027 (Appointed)*
Purdue University, West Lafayette, IN, USA

Postdoctoral Scholar *September 2026 - May 2027 (Appointed)*
University of California, Santa Cruz, CA, USA

Graduate Student Researcher *October 2021 - Current*
University of California, Santa Cruz, CA, USA

- Studied the security of intelligent cyber-physical systems and published the results in top conferences
- Developed an attack against Large Visual Language Models (LVLM) in embodied AI
- Used high-fidelity simulators to study the security of autonomous driving systems
- Implemented attack response strategies on robotic vehicles with Robot Operating System (ROS)

Graduate Research Assistant *February 2019 - July 2020*
Universidad de los Andes, Colombia

- Developed a strategy to detect and mitigate attacks against sensors of ground robots in a formation
- Developed sensor fusion and communication among ground robotic vehicles in a formation and published papers sharing the results

TEACHING EXPERIENCE

Teaching Assistant *Spring 2024*

University of California, Santa Cruz, CA, USA

- Discrete mathematics (CSE16, Undergraduate)

Graduate Teaching Assistant *August 2017 - December 2018*

Universidad de los Andes, Colombia

- Control Systems Analysis (IELE2300, Undergraduate)
- Dynamical Systems (IELE1502, Undergraduate)

Undergraduate Class Assistant *January 2017 - May 2017*

Universidad de los Andes, Colombia

AWARDS AND FELLOWSHIPS

- UCSC CSE Excellent Ph.D. Award. Santa Cruz, California, USA *June 2026*
- Best poster award, 1st Genzero Workshop. Abu Dhabi, United Arab Emirates *November 2024*
- Selected as a visiting scholar for the ELLIIT Focus Period. Lund, Sweden *April 2024*
- Travel award for VehicleSec 2024. San Diego, CA, USA *February 2024*
- Selected as a Cyber-Physical Systems Rising Star, NSF. Charlottesville, VA, USA *May 2023*

PUBLICATIONS

- D. Ortiz, E. Badal, O. Chang, **L. Burbano**, Z. Lei, E. Ka, S. Ukkusuri, L. H. Gilpin, A. A. Cardenas, "*Beyond Attacker-Defender Ratios in Counter-Swarm Defense*", Under review, 2026.
- Q. Sun, A. Abdo, **L. Burbano**, Z. Li, Y. Yao, A. A. Cardenas, Y. Cao. "*Beyond Crash: Hijacking Your Autonomous Vehicle for Fun and Profit*". Under review, 2026.
- **L. Burbano**, H. Sasahara, R. Song, Z. B. Celik, A. A. Cardenas. "*BADControl: Backdoor Attacks against Control Systems*". USENIX Security, 2026.
- **L. Burbano**, D. Ortiz, Q. Sun, S. Yang, C. Xie, Y. Cao, A. A. Cardenas. "*CHAI: Command Hijacking against Embodied AI*". Conference on Secure and Trustworthy Machine Learning (SaTML), 2026.
- D. Ortiz, M. Agrawal, Y. Malegaonkar, **L. Burbano**, A. Andersson, G. Dan, H. Sandberg, and A. A. Cardenas. "*Drones that Think on their Feet: Sudden Landing Decisions with Embodied AI*". preprint arXiv:2510.00167, 2025.
- D. Ortiz, **L. Burbano**, C. Hernandez, Z. Lei, Y. Park, S. Ukkusuri, and A. A. Cardenas. "*D4+: Emergent Adversarial Driving Maneuvers with Approximate Functional Optimization*". DDDAS Handbook Volume 4, 2025.
- **L. Burbano**, H. Sasahara, and A. A. Cardenas. "*Steerability of Autonomous Cyber-Defense Agents by Meta-Attackers*". Adaptive Cyber-Defense workshop at the Conference of Artificial Intelligence, 2025.
- S. Yang, Z. Wang, D. Ortiz, **L. Burbano**, M. Kantarcioglu, A. A. Cardenas, and C. Xie. "*Probing Vulnerabilities of Vision-LiDAR Based Autonomous Driving Systems*". Conference on Computer Vision and Pattern Recognition Workshops, 2025.
- D. Ortiz, **L. Burbano**, S. Yang, Z. Wang, A. A. Cardenas, C. Xie, and Y. Cao. "*Robust and Efficient AI-Based Attack Recovery in Autonomous Drones*". 1st Genzero Workshop, 2024.

- C. Hernandez*, D. Ortiz*, Z. Lei, **L. Burbano**, Y. Park, S. Ukkusuri, A. A. Cardenas "*D4: Dynamic Data-Driven Discovery of Adversarial Vehicle Maneuvers*". International Conference on Dynamic Data Driven Applications Systems, 2024. *equally contributed.
- L. Zhang*, **L. Burbano***, X. Chen, A. A. Cardenas, S. Drager, M. Anderson, and F. Kong. "*Fast Attack Recovery for Stochastic Cyber-Physical Systems*". IEEE Real-Time and Embedded Technology and Applications Symposium (RTAS), 2024. *equally contributed.
- **L. Burbano**, K. Garg, S. J. Leudo, A. A. Cardenas, and R. G. Sanfelice. "*Online Attack Recovery in Cyber-Physical Systems*". IEEE Security & Privacy Magazine, 2023.
- A. Alexandru, **L. Burbano**, M. F. Çeliktuğ, J. Gomez, A. A. Cardenas, M. Kantarcioglu, and J. Katz, "*Private Anomaly Detection in Linear Controllers: Garbled Circuits vs. Homomorphic Encryption*". IEEE Conference on Decision and Control (CDC), 2022.
- A. Zambrano, A. Palacio-Betancur, **L. Burbano**, A. F. Niño, L. F. Giraldo, M. Gutierrez-Soto, J. Giraldo, and A. A. Cardenas, "*You Make Me Tremble: A First Look at Attacks Against Structural Control Systems*". ACM SIGSAC Conference on Computer and Communications Security (CCS), 2021.
- **L. Burbano**, L. F. Cómbita, N. Quijano, and S. Rueda. "*Dynamic Data Integration for Resilience to Sensor Attacks in Multi-Agent Systems*". IEEE Access, 2021.
- G. Díaz-García, **L. Burbano**, N. Quijano, and L. F. Giraldo. "*Distributed MPC and Potential Game Controller for Consensus in Multiple Differential-Drive Robots*". IEEE Colombian Conference on Automatic Control (CCAC), 2019.
- **L. Burbano**, J. Lopez-Jimenez, and N. Quijano. "*Cosimulation tools for control in agricultural processes*". IEEE Colombian Conference on Automatic Control (CCAC), 2017.

TALKS

Workshop & Conference Talks

- "*CHAI: Command Hijacking against Embodied AI*",
  Hot Topics in the Science of Security Symposium (HotSoS), 2026.
- "*Steerability of Autonomous Cyber-Defense Agents by Meta-Attackers*",
  Hot Topics in the Science of Security Symposium (HotSoS), 2025.
- "*Private Anomaly Detection in Linear Controllers: Garbled Circuits vs. Homomorphic Encryption*",
  4th Norcal Control Workshop at the University of California, Santa Cruz. 2022

Invited Talks

- "*Robust and Efficient AI-Based Attack Recovery in Autonomous Drones*",
  Khalifa University, Abu Dhabi, United Arab Emirates. 2024
- "*Attack Response in Cyber-Physical Systems*",
  ACC'24 Student Workshop: Security and Privacy of the Next-Generation Cyber-Physical Systems, Toronto, Canada. 2024
- "*Fast Attack Recovery for Stochastic Cyber-Physical Systems*",
  Chalmers University of Technology, Gothenburg, Sweden. 2024

SERVICE

Volunteering

- Volunteered for UCSC's Mathematics, Engineering, Science Achievement (MESA) program, where I helped mentor students of underrepresented groups.

Mentored students

- Arthur Kraft. Visiting master's student from TU Berlin, 2026.
- Ariena Liang, Elina Fan, Sean Coleman. UCSC Cosmos, 2025.
- Carlos Hernandez. Student from San Jose State University, 2024. (Published one paper)

Reviewer

- PC member ACM Cyber-Physical System Security Workshop (CPSS) 2026.
- PC member Workshop on CPS & IoT Security and Privacy (CPSIoTSec) 2025.

SKILLS

- **Engineering Tools**: MATLAB/Simulink, LabVIEW, Latex, Robotic Operating System (ROS), Grafcet, CARLA, Gazebo, Airsim.
- **Programming Languages**: Python, Java, Ladder, C++.